54. We can try to infer the thought of Democritus from the discussions of Aristotle (for example, in *Physics* IV.213), but the evidence seems insufficient to me. See *Democrito. Raccolta dei frammenti, interpretazione e commentario di Salomon Luria* (Milan: Bompiani, 2007).

55. Unless the de Broglie-Bohm theory is true, in which case it has it—but hides it from us. Which is perhaps not so different in the end.

56. Carlo Rovelli, "Relational Quantum Mechanics," *International Journal of Theoretical Physics* 35 (1996): 1637, http://arxiv.org/abs/quant-ph/9609002. See also: "The Sky Is Blue and Birds Fly Through It," http://arxiv.org/abs/1712.02894.

57. Grateful Dead, "Walk in the Sunshine."

6. THE WORLD IS MADE OF EVENTS, NOT THINGS

58. Nelson Goodman, *The Structure of Appearance* (Cambridge, MA: Harvard University Press, 1951).

7. THE INADEQUACY OF GRAMMAR

59. For opposing views, see note 34.

60. In the terminology of a celebrated article by John McTaggart ("The Unreality of Time," *Mind*, N.S. 17 (1908): 457–74; reprinted in *The Philosophy of Time*, Le Poidevin and MacBeath), this is equivalent to denying the reality of the A-series (the organization of time into "past-present-future"). The meaning of temporal determinations would then be reduced to only the B-series (the organization of time into "before-it, after-it"). For McTaggart, this implies denying the reality of time. To my mind, McTaggart is too inflexible: the fact that my car works differently from how I'd imagined it and how I'd originally defined it in my head does not mean that my car is not real.

61. Letter by Einstein to the son and sister of Michele Besso, March 1955, in Albert Einstein and Michele Besso, *Correspondence, 1903–1955* (Paris: Hermann, 1972).

62. The classic argument for the block universe is given by the philosopher Hilary Putnam in a famous article published in 1967 ("Time and Physical Geometry," *Journal of Philosophy* 64 (1967): 240–47). Putnam uses Einstein's definition of simultaneity. As we have seen in note 27, if the Earth and *Proxima b* move with respect to one another, say they are approaching each other, an event A on Earth is simultaneous (for an earthling) to an event B on *Proxima b*, which in turn is simultaneous (for those on *Proxima b*) to an event C on Earth, *that is in the future of A*. Putnam assumes that "being simultaneous" implies "being real now," and deduces that the event in the future (such as C) is real now. The error is to assume that Einstein's definition of simultaneity has an ontological value, whereas it is only a definition of convenience. It serves to identify a relativistic notion that may be reduced to the nonrelativistic one through an approximation. But nonrelativistic simultaneity is a notion that is reflexive and transitive, whereas Einstein's is not, hence it makes no sense to assume that the two have the same ontological meaning beyond the approximation.

63. That the discovery by physics of the impossibility of presentism implies that time is illusory is an argument put forward by Gödel in "A Remark about the Relationship between Relativity Theory and Idealistic Philosophy," in *Albert Einstein: Philosopher-Scientist*, ed. P. A. Schlipp (Evanston, IL: Library of Living Philosophers, 1949). The error always lies in defining time as a single conceptual block that is either all there or not there at all. The point is discussed lucidly by Dorato, *Che cos'è il tempo?*, p. 77.

64. See, for instance, W. V. O. Quine, "On What There Is," *Review of Metaphysics* 2 (1948): 21–38, and the fine discussion of the meaning of reality in J. L. Austin, *Sense and Sensibilia* (Oxford: Clarendon Press, 1962).